

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 625 Artificial Intelligence — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Russell & Norvig — Artificial Intelligence: A Modern Approach, 4th edition	The primary source for this course. The site hosts free pseudocode for every algorithm, free Python implementations, and the figures. Library copy for the text; everything algorithmic is free.
Berkeley CS188 — Introduction to Artificial Intelligence (free lectures, notes, and the Pacman projects)	The best free AI course available, and the Pacman assignments are outstanding. Modules 02–05 and 10 follow it closely.
Millington — AI for Games, 3rd edition	The reference for Modules 04 and 11 — the engineering detail that academic treatments omit. The free <i>Game AI Pro</i> volumes at this link are chapters by shipping practitioners and are excellent.
Ghallab, Nau & Traverso — Automated Planning and Acting (free draft)	Module 07's reference , with the relaxation-based heuristics developed properly.
Biere, Heule, van Maaren & Walsh — Handbook of Satisfiability (chapters free)	Module 08's CDCL material , and the competition results that show how far solvers have come.
Sutton & Barto — Reinforcement Learning, 2nd edition (free PDF)	Chapters 3 and 4 are Module 10 , and the rest is CSCE 642 — so the two courses share one free book, which makes the connection easy to follow.

Module-by-module map

Module 01 — Agents and Rationality

Resource	What it covers for this module
Russell & Norvig — AIMA, chapters 1–2 (pseudocode and code free)	The agent and environment definitions of §1 and §2 , in their standard form.
Berkeley CS188 — lecture 1 and the course overview (free)	The same material with better motivation , and the Pacman project framework you will use from Module 02.
McCorduck — Machines Who Think	The history behind §3's AI effect , written by someone who interviewed the people involved. Library copy.
Game AI Pro — the free volumes (free)	Chapters by shipping practitioners , and the clearest evidence for §4's argument about what production systems actually use.

Module 02 — Search

Resource	What it covers for this module
Russell & Norvig — AIMA, chapter 3 (pseudocode free)	The reference for this module. The generic algorithm of §1 and the comparison table of §2 are from here.
Berkeley CS188 — search lectures and Project 1 (free)	The Pacman search project , which is the best available exercise for this material and includes the state-formulation decisions of §1.
Korf — Depth-First Iterative-Deepening (free)	The §2 result , with the node-count argument worked out properly.
Red Blob Games — Introduction to Graph Search (free, interactive)	Interactive visualisations of every strategy in §2 , and the clearest intuition-builder available.

Module 03 — Heuristic Search

Resource	What it covers for this module
Russell & Norvig — AIMA, chapter 3 (sections on informed search)	A*, admissibility, consistency, and the relaxation argument of §3 , with the 8-puzzle worked through.
Hart, Nilsson & Raphael — A Formal Basis for the Heuristic Determination of Minimum Cost Paths (free)	The original A* paper , with the optimality proof. Short and readable.
Korf — Disjoint Pattern Database Heuristics (free)	The §3 pattern database method , including the additivity condition that makes them so strong.
Red Blob Games — Heuristics (free, interactive)	The practical guide for grid heuristics , including weighted A* and tie-breaking, with interactive demonstrations.

Module 04 — Pathfinding in Practice

Resource	What it covers for this module
Recast & Detour (free, open source)	The navigation mesh builder of §1 , used in many shipped games. Readable, and worth reading for the voxelisation pipeline.
Botea, Müller & Schaeffer — Near Optimal Hierarchical Path-Finding (HPA*) (free)	The §3 hierarchy , with the node-count reductions measured.
Koenig & Likhachev — D* Lite (free)	The §3 replanning method , and notably simpler than the original D*.
Red Blob Games — A* and pathfinding articles (free, interactive)	The best practical treatment of this module available , including grids, navmeshes, smoothing, and flow fields, all interactive.

Module 05 — Adversarial Search

Resource	What it covers for this module
Russell & Norvig — AIMA, chapter 5 (pseudocode free)	Minimax, alpha-beta, and the ordering results of §1 , with the complexity derivations.
Chess Programming Wiki (free)	The practical reference for §1 and §2 — move ordering, transposition tables, and quiescence search, written by engine authors.
Browne et al. — A Survey of Monte Carlo Tree Search Methods (free)	The §3 reference , with UCB1 derived and the variants catalogued.
Silver et al. — Mastering the Game of Go without Human Knowledge (free)	The synthesis of §3's table footnote — MCTS with learned evaluation and prior.

Module 06 — Constraint Satisfaction

Resource	What it covers for this module
Russell & Norvig — AIMA, chapter 6 (pseudocode free)	The reference for §1 through §3 , including AC-3 and the tree-structured result of §3.
Google OR-Tools — CP-SAT documentation (free)	A production constraint solver , with the global constraints of §2 and worked examples of every problem in §1's table.
Gumin — Wave Function Collapse (free, open source)	The §4 algorithm , with the original implementation and a large gallery of results.
Short & Adams (eds.) — Procedural Generation in Game Design	The §4 design perspective , by practitioners. Library copy; the free PCG Book at procedural-generation sites covers similar ground.

Module 07 — Classical Planning

Resource	What it covers for this module
Russell & Norvig — AIMA, chapters 11–12 (pseudocode free)	STRIPS, the delete relaxation, and the heuristics of §2 , developed carefully.

Resource	What it covers for this module
Hoffmann & Nebel — The FF Planning System (free)	The h_FF heuristic and helpful actions of §2 , which is the paper that changed the field's capability.
Orkin — Three States and a Plan: The AI of F.E.A.R. (free)	The §3 GOAP case study , by the person who shipped it. The clearest account of both the appeal and the cost.
Ghallab, Nau & Traverso — Automated Planning and Acting (free draft)	The reference for §4 , including HTN planning and the treatment of acting under the violated assumptions.

Module 08 — Logic and Satisfiability

Resource	What it covers for this module
Russell & Norvig — AIMA, chapters 7–9 (pseudocode free)	Propositional and first-order logic, resolution, and DPLL — the reference for §1, §2, and §4.
Marques-Silva, Lynce & Malik — Conflict-Driven Clause Learning (free chapter)	The §2 mechanism , by its originators, with the implication-graph analysis worked through.
Z3 and MiniSat documentation and tutorials (free)	The practical tools of §3 . The Z3 guide is interactive and is the fastest way to become useful with SMT.
Biere, Heule, van Maaren & Walsh — Handbook of Satisfiability	The comprehensive reference , with several chapters free and the annual competition results that document §2's progress.

Module 09 — Probabilistic Reasoning

Resource	What it covers for this module
Russell & Norvig — AIMA, chapters 12–14 (pseudocode free)	Bayes nets, d-separation, and variable elimination — the reference for the whole module.
Koller & Friedman — Probabilistic Graphical Models	The comprehensive treatment , strong on the treewidth connection of §3. Library copy; the author's free Coursera lectures cover much of it.
Pearl — Causality, chapter 1; and The Book of Why	Where §2's d-separation and explaining away come from , and the causal reading of the graph that the standard treatment understates.
pgmpy documentation (free)	A working library for §2 and §3 , with every inference method in the table — after implementing variable elimination yourself once.

Module 10 — Decisions Under Uncertainty

Resource	What it covers for this module
Sutton & Barto — Reinforcement Learning, chapters 3–4 (free PDF)	MDPs, the Bellman equation, and dynamic programming — the reference for this module, and the rest of the book is CSCE 642.

Resource	What it covers for this module
Berkeley CS188 — MDP and reinforcement learning lectures (free)	§1 and §2 with excellent worked examples , and a project implementing both algorithms.
Kaelbling, Littman & Cassandra — Planning and Acting in Partially Observable Stochastic Domains (free)	The §3 reference , and the clearest statement of why belief space is the right formulation.
Kurniawati, Hsu & Lee — SARSOP (free)	A practical POMDP solver , which makes §3's approximations concrete.

Module 11 — Game AI Architecture

Resource	What it covers for this module
Game AI Pro, volumes 1–3 (free online)	The reference for this entire module — chapters by shipping practitioners on behaviour trees, utility systems, steering, and the frame-budget engineering of §1.
Millington — AI for Games, 3rd edition	The systematic treatment of §2 and §3 , with pseudocode throughout. Library copy.
Colledanchise & Ögren — Behavior Trees in Robotics and AI (free PDF)	§2 treated formally , including the relationship to state machines and the composability argument made precise.
Reynolds — Steering Behaviors For Autonomous Characters (free)	The §3 behaviours, from the originator , with demonstrations. The flocking paper is here too.

Module 12 — Multiple Agents

Resource	What it covers for this module
Sharon et al. — Conflict-Based Search for Optimal Multi-Agent Pathfinding (free)	The §2 algorithm , with the branching argument and the comparison against coupled search.
van den Berg et al. — Reciprocal n-body Collision Avoidance (ORCA) (free)	The §1 and §3 method , with a free reference implementation used widely in games.
Reynolds — Flocks, Herds, and Schools (free)	The §3 emergence example , and the paper is candid about the rules being found by iteration.
Shoham & Leyton-Brown — Multiagent Systems (free PDF)	Free in full, and the reference for §4 — game theory, coalitions, and mechanism design, aimed at computer scientists.

Module 13 — Choosing an Approach

Resource	What it covers for this module
Russell & Norvig — AIMA, chapter 1 and the concluding chapter	The field's own account of the symbolic/learned division of §2 , from a source that covers both.

Resource	What it covers for this module
Game AI Pro — the architecture and postmortem chapters (free)	§1's table as practitioners apply it , with the constraints that drove each decision stated.
Sculley et al. — Hidden Technical Debt in Machine Learning Systems (free)	Why the hybrid interface of §2 is where the maintenance cost accumulates — relevant well beyond machine learning.
Berkeley CS188 — the full project set (free)	If you want more practice , the five Pacman projects cover Modules 02, 03, 05, 06, 09, and 10 with automatic grading.